

Generation Interconnection System Impact Study Report

For Queue Project Q15-0496

Boggy Brook 138 kV

October 20, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0496. The Interconnection Customer proposes a 105.9 MW solar photovoltaic facility interconnecting at the Boggy Brook 138 kV substation in the Pennsylvania area. The request was received on July 04, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 259 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$22,595,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Chilliwick – Sea Island 138 kV Line	Super Max Substation – Green Country 138 kV	158.3	306	20.1
Super Max – Chase 138 kV	Chilliwick – 26Th And F 138 kV line	155.1	234	14.8
26Th And F – Chilliwick 138 kV	Super Max – Green Country 138 kV line	153.8	337	20.1
Chichester – 26Th And F 138 kV	26Th And F 138/1kV TX	141.8	203	23.9
Green Country – Super Max 138 kV Line	Chilliwick – 26Th And F 138 kV	138.3	272	14.8
Chichester – Lake Cowichan 138 kV	26Th And F 138 kV/1 kV Transformer Bank	136.7	219	23.9
6Th St – Sea Island 138 kV	Sea Island Substation – Beechhust 138 kV Line	131.2	234	14.7
Warehouse Electric – Liberty Electric Power Plant 138 kV line	6Th St – Sea Island 138 kV	129.1	246	8.5
Sea Island – Beechurst 138 kV Line	Sea Island – 6Th Street 138 kV L	117.8	261	13.3
Green Country – Red Rock 138 kV	Chichester Substation – Lake Cowichn 138 kV line	116.6	224	19.1
Protein – 26Th And F 138 kV Line	26Th And F – Chilliwick 138 kV	110.0	243	5.3
Fall Creek – Protein 138 kV	26Th And F – Chilliwick 138 kV L	100.3	224	5.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Chilliwick – Sea Island 138 kV Line	Rebuild line with higher-capacity conductor	\$4,752,000	64% (\$3,017,000)
Super Max – Chase 138 kV	Rebuild line with higher-capacity conductor	\$1,443,000	64% (\$921,000)
26Th And F – Chilliwick 138 kV	Rebuild line with higher-capacity conductor	\$1,314,000	64% (\$834,000)
Chichester – 26Th And F 138 kV line	Rebuild line with higher-capacity conductor	\$6,355,000	100% (\$6,355,000)
Super Max – Green Country 138 kV	Rebuild line with higher-capacity conductor	\$6,690,000	64% (\$4,271,000)
Chichester – Lake Cowichan 138 kV	Rebuild line with higher-capacity conductor	\$1,282,000	100% (\$1,282,000)
6Th St – Sea Island 138 kV	Rebuild line with higher-capacity conductor	\$2,731,000	62% (\$1,705,000)
Liberty Electric Power Plant – Warehouse Electric 138 kV	Rebuild line with higher-capacity conductor	\$1,313,000	44% (\$576,000)
Sea Island – Beechhurst 138 kV	Rebuild line with higher-capacity conductor	\$1,118,000	64% (\$717,000)
Red Rock – Green Country 138 kV	Rebuild line with higher-capacity conductor	\$1,039,000	100% (\$1,039,000)
Protein – 26Th And F 138 kV Line	Rebuild line with higher-capacity conductor	\$1,472,000	100% (\$1,472,000)
Fall Creek – Protein 138 kV	Reconductor line	\$406,000	100% (\$406,000)
	Total allocated to project		\$22,595,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.